Variable Transformations

STAT 24510-30040

(Winter 2025)

We are often in the need of knowing the behavior of functions of random variables. The techniques of variable transformation methods allow us to obtain information and properties of functions of random variable (or random vector) X from the probability distribution properties of the original X.

Many common random variables are functions of other random variables. For example, common statistic χ^2, t, and F are built as functions of normal random variables.

In this section, we state commonly used formulas for variable transformations obtained from calculus, focusing on probability density of functions of random variables.

Change of variables — Scalars

In the case of one-dimensional, univariate random variables, if a random variable Y is defined as a function of another random variable X,

$$Y = g(X)$$

what is the probability distribution of Y in terms of the probability distribution of X?

The relationship also depends on the properties of the function $g(x)$.

The fundamental relation of the probability distribution of Y in terms of X is

$$\mathbb{P}(Y \in A) = \mathbb{P}\left(g(X) \in A\right) \tag{1}$$

where A is any (measurable) event. In the univariate scalar case of $X, Y \in \mathbb{R}$, event $A \subset \mathbb{R}$.
Often A is an interval or a union of finite or of countably many intervals.

The inverse function of g may not exist. The inverse mapping of g is defined as

$$g^{-1}(A) = \{x \in \mathbb{R} : \ g(x) \in A\} \tag{2}$$

which is well defined for reasonable g (e.g. piecewise monotone) and most events (measurable set A).

Then we can write

$$\mathbb{P}(Y \in A) = \mathbb{P}\left(X \in g^{-1}(A)\right), \qquad A \subset \mathbb{R}$$

This defines the probability distribution of Y in terms of the probability distribution of X, as desired.

For example, if $y = g(x) = x^2$, $A = \{y_o\} = \{4\}$ consists of one point, then

$$g^{-1}(A) = g^{-1}(\{4\}) = \{x \in \mathbb{R} : \ x^2 = 4\} = \{x : x = -2, x = 2\}$$

The set on the right hand side contains all x such that $g(x) = y_o = 4$. If g is a one-to-one function, such as monotone functions, then inverse function g^{-1} exists, and the inverse $x = g^{-1}(y)$ is unique.

Functions of discrete random variables

If X is a discrete random variable, then $Y = g(X)$ is also discrete. It is often easier to calculate the probability mass function of Y directly,

$$\mathbb{P}(Y = y) = \sum_{\{x:\ g(x)=y\}} \mathbb{P}(X = x)$$

Functions of continuous random variables

If X is a discrete random variable, then $Y = g(X)$ can be discrete, continuous, or mixed.

When X and $Y = g(X)$ both are continuous random variables, the probability equation can be expressed in terms of integrations of their density functions f_X and f_Y.

$$\mathbb{P}(Y \in A) = \int_A f_Y(y)dy = \int_{g^{-1}(A)} f_X(x)dx = \mathbb{P}\left(X \in g^{-1}(A)\right), \qquad A \subset \mathbb{R}$$

Monotone transformation for continuous random variables

Assume X and $Y = g(X)$ are continuous random variables with density functions $f_X(x)$ and $f_Y(y)$ respectively. Recall the relation of derivatives of inverse functions from calculus,

$$\frac{d}{dy} g^{-1}(y) = \frac{1}{g'(g^{-1}(y))}$$

when both sides are well defined.

When g is a strictly increasing function with $g'(x) > 0$,

$$F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}(g(X) \le y) = \mathbb{P}(X \le g^{-1}(y)) = F_X(g^{-1}(y))$$

Taking derivative with respect to y,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X(g^{-1}(y)) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

When g is a strictly decreasing function with $g'(x) < 0$, then

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \le y) = \mathbb{P}(g(X) \le y) \\ &= \mathbb{P}(X \ge g^{-1}(y)) = 1 - \mathbb{P}(X < g^{-1}(y)) \\ &= 1 - \mathbb{P}(X \le g^{-1}(y)) = 1 - F_X(g^{-1}(y)) \end{aligned}$$

Taking derivative with respect to y,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = -\frac{d}{dy} F_X(g^{-1}(y)) = f_X(g^{-1}(y)) \left(-\frac{d}{dy} g^{-1}(y) \right) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Therefore when g is a monotone function with continuous derivative, the relation between the density functions can be summarized as

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| \tag{3}$$

when the derivative of $g^{-1}(y)$ exists. An alternative expression can be

$$f_Y(y) = f_X(x)\left|\frac{1}{g'(x)}\right|, \qquad x = g^{-1}(y), \tag{4}$$

when $g'(x)$ is non-zero. The relation can be loosely represented using the symbolic differential notations as

$$f_Y(y)dy = f_X(x)dx$$

combined with

$$\frac{dy}{dx} = g'(x), \qquad \frac{dx}{dy} = \left[g'(x)\right]^{-1} = \left[g'(g^{-1}(y))\right]^{-1}$$

Remarks

The above relation and formulas apply when all terms are well defined. The assumptions and formulas serve as guidelines, they are not In applications there are adjustments according to the context and based on the fundamental relation $\mathbb{P}(Y \in A) = \mathbb{P}\{g(X) \in A\}$. The following are some common adjustments.

- The relation $Y = g(X)$ and $|g'(x)| > 0$ may hold only on an interval I, and $f(x) = 0$ for $x \notin I$.

  Then $f_Y(y) = 0$ for $y = g(x), x \notin I$.

- For discrete random variables,

  $$F(x) = \mathbb{P}(X \le x) = \sum_{k \le x} \mathbb{P}(X = k)$$

  thus can be expressed as a sum of probability mass function over $(-\infty, x]$. Note that for discrete random variables,

  $$\mathbb{P}(X < x) \ne \mathbb{P}(X \le x), \qquad when \quad \mathbb{P}(X = x) > 0$$

- The requirement of existence of g' and even monotonicity of g are for their convenience and commonality. They are sufficient conditions, not necessary ones.

  In specific problems, especially, when X and Y have a clear one-to-one relationship, the relationship of their densities or mass functions can be derived directly, both smoothness condition and monotonicity condition are not necessary.

Example

$X > 0$ is a continuous random variable with density function $f_X(x)$. Define

$$Y = X^3$$

This is the simple case that $g(x) = x^3$ is a strictly increasing function.

Then for $x > 0$,

$$y = g(x) = x^3, \quad g'(x) = 3x^2, \quad x = y^{1/3} = g^{-1}(y)$$

and

$$f_Y(y) = f_X(x)\left|\frac{1}{g'(x)}\right| = \frac{f_X(x)}{3x^2} = \frac{f_X\left(y^{1/3}\right)}{3y^{2/3}}$$

In this case $y >$ if and only if $x > 0$. Thus

$$f_Y(y) = \begin{cases} \frac{f_X\left(y^{1/3}\right)}{3y^{2/3}}, & y > 0, \\ 0, & y \le 0. \end{cases}$$

Example

$X > 0$ is a continuous random variable with density function $f_X(x)$. Define

$$Y = 1/X$$

This is the case that $g(x) = 1/x$ is a strictly decreasing function.

Then for $x > 0$,

$$y = g(x) = 1/x, \quad g'(x) = -1/x^2, \quad x = 1/y = g^{-1}(y)$$

The density of Y is

$$f_Y(y) = f_X(x)\left|\frac{1}{g'(x)}\right| = \frac{f_X(x)}{|-1/x^2|} = \frac{f_X\left(1/y\right)}{y^2}$$

As in the previous example, $y >$ if and only if $x > 0$.

Therefore

$$f_Y(y) = \begin{cases} \frac{f_X(1/y)}{y^2}, & y > 0, \\ 0, & y \le 0. \end{cases}$$

Since $g' < 0$, integral limits needs to be carefully inverted.

Piecewise monotone transformation

If $y = g(x)$ is a piecewise monotone function, then the formulas (3) or (4) that relate the density functions in strictly monotone case can be applied to each monotone piece.

For each y value, the value of the corresponding density function needs to add all pieces containing an x with $g(x) = y$. The density function of Y has the expression

$$f_Y(y) = \sum_{\{x: g(x) = y\}} f_X(x)\left|\frac{1}{g'(x)}\right| \tag{5}$$

For each value of y, the summation should be understood as adding over all x such that $x = g(y)$, that is, all x in the inverse mapping set

$$g^{-1}(\{y\}) = \{x : g(x) = y\}$$

Formula (5) is equivalent to applying (4)

$$f_Y(y) = f_X(x)\left|\frac{1}{g'(x)}\right|, \qquad x = g^{-1}(y)$$

on each interval I_i where $y = g(x)$ is a monotone function, then sum the density over all such intervals.

Example

$X \in \mathbb{R}$ is a continuous random variable with density function $f_X(x), x \in \mathbb{R}$.

$$Y = X^2$$

We have $g(x) = x^2, g'(x) = 2x$. Each y value corresponds to two x values $x = \pm\sqrt{y}$. The density of Y is

$$f_Y(y) = \sum_{\{x: g(x) = y\}} \frac{f_X(x)}{|g'(x)|} = \sum_{\{x:\ x = \sqrt{y}\ or\ x = -\sqrt{y}\}} \frac{f_X(x)}{|2x|}$$

Expressed in terms of y,

$$f_Y(y) = \frac{f_X(\sqrt{y})}{|2\sqrt{y}|} + \frac{f_X(-\sqrt{y})}{|-2\sqrt{y}|} = \frac{1}{2\sqrt{y}}\left[f_X(\sqrt{y}) + f_X(-\sqrt{y})\right]$$

Example

X is a continuous random variable with density $f_X(x)$. Define

$$Y = \sin X$$

Then the density of Y consists of an infinite sum

$$f_Y(y) = \sum_{\{x:\ \sin x = y\}} \frac{f_X(x)}{|\cos x|}$$

where the summation is over all x in the set

$$\{x:\ \sin x = y\} = \{x:\ x = \arcsin y + 2k\pi \ \text{ or } \ x = \pi - \arcsin y + 2k\pi,\ \forall k = 0, \pm1, \pm2, \cdots\}$$

Finally, the density of Y should be expressed in terms of y.

Change of variables — Vectors

In many applications, we need to expression a set of random variables $\{X_1, \cdots, X_p\}$ in terms of another set of random variables $\{Y_1, \cdots, Y_k\}$, where each random variable Y_j is a function of $\{X_1, \cdots, X_p\}$.

The most common function forms of statistics are sums, products (including powers), and quotients.

In the following we consider the two-dimensional vector-to-vector case $p = k = 2$.

Suppose random vector (X, Y) has joint density function $f_{X,Y}(x, y)$. Let random vector (U, V) be defined as

$$\begin{cases} U = u(X, Y) \\ V = v(X, Y) \end{cases}$$

What is the joint density function of the random vector (U, V)?

As in univariate case, the fundamental relation of the joint probability distribution of (U, V) in terms of (X, Y) is

$$\mathbb{P}\{(U, V) \in B\} = \mathbb{P}\{(X, Y) \in A\}, \qquad A = \{(x, y) : (u(x, y), v(x, y)) \in B\}$$

For discrete random variables, only countably many (x, y) with positive probability, the corresponding (u, v) with positive probabilities identified.

For continuous random variables with density functions,

$$\iint_{\mathbb{R}^2} f_{U,V}(u, v)\,du\,dv = \iint_{\mathbb{R}^2} f_{X,Y}(x, y)\,dx\,dy = 1.$$

If the transformation is one-to-one locally, the relationship of density functions under change of variables can be symbolized as

$$f_{U,V}(u, v)\,du\,dv = f_{X,Y}(x, y)\,dx\,dy$$

Consider the case of one-to-one transformations, that each (u, v) maps to unique (x, y) and vice versa. Then we can invert the relation to express (X, Y) in terms of (U, V).

When u, v have continuous partial derivatives with respect (x, y), The joint density functions have the relation

$$f_{X,Y}(x, y) = f_{U,V}(u, v)\,|J|$$

where J is the "Jacobian" or "Jacobian determinant". The partial derivation matrix of functions (u, v) with respect to (x, y) is

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{bmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\[2ex] \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{bmatrix}$$

The Jacobian J is the determinant of the partial derivation matrix,

$$J \ = \ det\left(\frac{\partial(u, v)}{\partial(x, y)}\right) \ = \ \left|\frac{\partial(u, v)}{\partial(x, y)}\right| \ = \ \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\[2ex] \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix}$$

The Jacobian can be viewed as the rate of change in the infinitesimal volumes. It carries essential information about the behavior of the transformation function (u, v) near a point.

If (u, v) are continuously differentiable functions of x and y, and if $J \neq 0$ at a point, then in the neighborhood of the point, (x, y) can be written as functions of (u, v) (the inverse function theorem).

When $J \neq 0$, we may take partial derivatives of (x, y) with respect to variables (u, v),

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{bmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\[2ex] \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{bmatrix}$$

The corresponding Jacobian is

$$J^* \ = \ \left|\frac{\partial(x, y)}{\partial(u, v)}\right| \ = \ \left|\frac{\partial(u, v)}{\partial(x, y)}\right|^{-1}$$

So, in fact,

$$J^* = J^{-1} = \frac{1}{J}$$

The left most equality above is meant to be the definition of the notation J^{-1} as the Jacobian of the inverse functions, while the last equality serves as a justification of the appropriateness of the notation.

So we can define the Jacobian and inverse Jacobian interchangeably.

When all partial derivatives are smooth, there is a stronger inverse matrix relation

$$\begin{bmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\[2ex] \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{bmatrix} = \text{Inverse matrix of} \begin{bmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\[2ex] \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\[2ex] \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{bmatrix}^{-1}$$

where the last expression is the notation for matrix inverse.

The equality can be verified by matrix multiplication and repeated use of the chain rule in differentiation, under the assumption that the Jacobian is non-zero at the point.

Example (Sum of two random variables)

Random variables X and Y has density and joint density functions $f_X(x), f_Y(y)$ and $f_{X,Y}(x,y)$ respectively.
Find the density function of $X+Y$ (using 2-dimensional variable transformation).

Define U, V as

$$\begin{cases} U = X+Y \\ V = Y \end{cases}$$

The Jacobian

$$J = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1$$

From the density conversion formula

$$f_{U,V}(u,v)|J| = f_{X,Y}(x,y)$$

we obtain the joint density for (U,V) in terms of the known joint density of (X,Y),

$$f_{U,V}(u,v) = f_{X,Y}(u-v,v)$$

The desired marginal density function for $U = X+Y$ can be obtained by integrating out v.

$$f_U(u) = \int_{\mathbb{R}} f_{U,V}(u,v)dv = \int_{\mathbb{R}} f_{X,Y}(u-v,v)dv$$

When X and Y are independent, the density of the sum can be expressed as a convolution product of the densities of X and Y.

$$f_U(u) = \int_{\mathbb{R}} f_X(u-v)f_Y(v)dv = f_X * f_Y$$

Note that in this case, the inverse functions exist and can be written as

$$\begin{cases} X = U-V \\ Y = V \end{cases}$$

Sometimes these inverse functions are easier to work with.

Example (Quotient of random variables)

Continuous random variables X and Y have marginal densities $f_X(x), f_Y(y)$ and joint density function $f_{X,Y}(x,y)$.
Objective: Find the density function of X/Y (using 2-dimensional variable transformation), assuming $Y \neq 0$.

Define U, V as

$$\begin{cases} U = X/Y \\ V = Y \end{cases}$$

We may consider using

$$f_{X,Y}(x,y) = f_{U,V}(u,v)\,|J|$$

and computing the Jacobian

$$J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

as in the previous example. However, in this case, J^{-1} is the easier one to calculate (though J is not hard either). We will use

$$f_{X,Y}(x,y)\,\left|J^{-1}\right| = f_{U,V}(u,v)$$

with

$$J^{-1} = \left|\frac{\partial(x,y)}{\partial(u,v)}\right|$$

Writing (X,Y) as function of (U,V),

$$\begin{cases} X = UV \\ Y = V \end{cases}$$

Then $\frac{\partial(x,y)}{\partial(u,v)}$ has simpler entries.

$$J^{-1} = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} v & u \\ 0 & 1 \end{vmatrix} = v$$

We obtain the joint density function

$$f_{U,V}(u,v) = f_{X,Y}(uv,v)|v|$$

To obtain the marginal density function of the quotient $U = X/Y$, we do

$$f_U(u) = \int_{\mathbb{R}} f_{X,Y}(uv,v)|v|dv$$

When X and Y are independent, the quotient X/Y has a factorable density function

$$f_U(u) = \int_{\mathbb{R}} f_X(uv)f_Y(v)|v|dv$$

Notes Relevant sections in Rice: 2.3, 3.6 and various applications.